

T-740-SPMM Project B: Discovery

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Executive Summary

This report outlines the discovery phase of the project assigned to our team. The project aims to add a feature to the Ísland.is application that will allow Icelandic residents to securely access and present a medical summary while traveling abroad. The project's goal is to enhance patient safety and the quality of care in critical situations by giving travelers a concise medical history.

In collaboration with Stafrænt Ísland and key healthcare partners, the project team completed the discovery phase. This phase had five main goals:

- Stakeholder interviews to understand the project's vision, requirements, and expectations.
- Task Analysis by outlining the tasks and workflows that users need to complete to achieve their goals.
- User groups identification.
- User interviews to understand their needs, challenges, and expectations.
- UX goal definition.

The project started with a pre-analysis, where the team identified key user groups and created a preliminary project map. After a meeting with stakeholders, the team gained a clearer understanding of the project's vision and expectations, which led to a significant change in focus. The main takeaway from the meeting was the importance of prioritizing user benefits, practical use cases, and conceptual solutions over current technological limits. The stakeholders encouraged the team to explore different scenarios and highlighted crucial themes for development.

Moreover, the team conducted user interviews to better understand their needs and concerns. With these new insights, the team gathered to update their initial materials, modifying their pre-analysis to reflect the new findings from both the stakeholder meetings and user interviews. As the final step the team defined three core UX goals: security, relief and control.

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1 Introduction

When Icelandic residents need medical care abroad, having access to an accurate medical history can be extremely helpful. Project B examines the discovery phase of a potential extension to the Ísland.is application that would enable Icelandic citizens and residents to present a reliable medical summary while traveling. This initiative arose from a stakeholder sprint, which emphasized that providing a concise and accurate medical summary can greatly improve the quality of care and enhance patient safety abroad, particularly in critical situations.

Working with Stafrænt Ísland and other healthcare stakeholders, our team sets out to understand needs and constraints before any design or development. This report pursues five goals in the discovery phase: (1) analyze the current state of the project through stakeholder meetings; (2) map end-to-end usage scenarios for presenting a medical summary abroad; (3) identify and characterize relevant user groups; (4) conduct semi-structured interviews to gain insights into user needs and expectations ; (5) and define user-experience (UX) goals. To ensure a structured approach, the team applied templates and methods provided by *UCDSprint.com*, used to analyze user groups, map user journeys and define UX goals.

The remainder of this report will first cover the as-is analysis, after which the mapping of steps to reach the user goal is discussed. Next, the report mentions the user group analysis and the results of the stakeholder meeting. Finally, the interviews with users and UX goals are discussed.

2 Main

2.1 As-is analysis

For Icelandic travelers, there is currently no standardized way to carry medical summaries digitally abroad. Individuals often rely on printed documents, doctor's notes or MedicAlert IDs, if prepared in advance. This highlights a gap between having access to health records and receiving appropriate care when seeking medical attention abroad.

In collaboration with *Ísland.is*, there is interest in exploring an extension that would allow individuals to download and securely access their medical summary in times of need. The current vision remains fragmented: While infrastructure for online access is improving within Iceland and parts of Europe, people traveling more widely lack a reliable and user-controlled method to present their health information.

To move this concept forward, several challenges must be addressed: we must ensure that only relevant health information is shared, overcome language barriers, and establish trust in the authenticity of the records.

2.2 Mapping

The first step of Project B was mapping the end-to-end usage scenarios for presenting a medical summary abroad, which was done in a simple chart. First, we identified the main user groups for our product and defined the common goal. Next, they outlined the steps each user group would take to achieve this overarching goal. After that, we connected all steps with arrows, creating a clear visual of the user's path. This mapping gives a basic overview, helping us better understand the project's structure and user experiences.

Initially, we defined three distinct user groups: Icelandic travelers, healthcare professionals within Iceland, and healthcare professionals abroad. Icelandic travelers were considered the main users of our system, allowing them to download their medical records as a PDF, QR code, or digital wallet file. Moreover, Healthcare professionals within Iceland were considered responsible for ensuring data accuracy that is collected before traveling and maintaining contact with providers abroad, while healthcare professionals abroad were thought to be responsible for collecting patient data through summaries and providing care abroad. We also considered sending information from abroad back to Iceland, but it proved to be beyond the scope of this project. The overarching goal was defined as "having better medical care outside Iceland".

However, the stakeholder meeting revealed that our initial approach was too detailed. Therefore,

we decided, based on stakeholder wishes, to streamline our process map by eliminating "healthcare professionals within Iceland" and removing other tasks beyond our scope. This resulted in the following updated workflow: Travelers log in to the *Ísland.is* app to access their medical records before traveling. Within the app, they select the information they want to include in the summary. They can also choose a language for eventual translation and download an offline copy for easy access. This ensures that they have a translated medical summary readily available, even when Internet connection happens to be absent while traveling. When Internet is available, users can also translate the summary in real time. Once medical care is needed abroad, a traveler gives permission for a healthcare professional to receive and read their medical summary. This provides the professional with new translated information that could help them re-evaluate and treat the patient.

Furthermore, the target group was selected based on the user feedback and stakeholder meetings. The team collectively agreed that the target user group should be travelers, since these are the main user of our app, requiring the biggest share of tasks for usage. The focus was placed on the process from which users select their medical information to traveling with it (see the red circle in Appendix C.2). Finally, the common goal in the map was changed based on UX goals, which is elaborated further on in this report.

2.3 User group analysis

The user group analysis was conducted with a template from *UCDSprint.com* called "user group analysis template". This template was filled out and can be found in the appendix B. This section discusses the user groups according to template results.

Two user groups were defined for our system that provides a medical summary abroad: Icelandic travelers and healthcare professionals outside of Iceland (see appendix B). The first user group is Icelandic travelers, which can potentially consist of thousands of people. Icelandic travelers cover a broad demographic, ranging from children being accompanied by parents, to adults and elderly citizens who have an Icelandic personal identification number (*kennitala*, in Icelandic). Also, this user group can vary greatly in background of education and level of health literacy. Similarly, Icelandic travelers differ in digital literacy, with computer skills ranging from basic to advanced smartphone use. Moreover, this user group can contain individuals with chronic conditions, congenital conditions and disabilities in any form. The common denominator between individuals in this user group is their need to have simple access to a reliable medical summary while traveling abroad. The medical summary is accessed through *island.is* and usage occurs mainly during emergencies or medical encounters abroad. The users are expected to be able to open the app and provide access to their medical summary. The focus lies on users gaining better medical care abroad, ensuring their health history is taken into account.

The second user group is healthcare professionals outside of Iceland. This group represents a vast group of clinicians and healthcare professionals worldwide who could potentially treat Icelandic travelers. They are medically trained with the ability to read and understand their respective country's electronic health records. Their interaction with the medical summary system occurs abroad in clinics, hospitals, or emergency rooms when an Icelandic traveler requires care. Their main goal is to access a reliable summary of a patient's health records, which they are required to effectively interpret. For this group it is important to integrate previous medical information about patients into their treatment plan, ensuring better overall assessment and continuity of care.

Considering both user groups, the medical summary is designed to balance accessibility and usability for travelers while providing clarity, accuracy and reliability for the healthcare professionals who will provide care to Icelandic travelers. For both groups, the summaries ensure that Icelandic citizens and residents could receive safe, informed and better medical care wherever they are in the world.

2.4 Meeting with stakeholder

On the 16th of September at 9:00 AM, the team held a scheduled meeting with the stakeholders. The purpose of this meeting was to gain clarity on the stakeholders' overall vision for the project and to ensure alignment between the team's ideas and the stakeholders' expectations. The meeting was designed to be open and interactive, giving the team the opportunity to raise questions and address uncertainties that had been carried forward from previous discussions.

In preparation for this meeting, the team wrote down a set of key questions that had come up during earlier project conversations. These questions were intended to provide deeper understanding on the stakeholders' long term vision and to resolve any ambiguities concerning the scope, process and intended outcome. Specifically, the team sought to better understand the progress the stakeholders had made so far, their perspective on technical and design choices, such as on-the-fly translation, and their expectations regarding privacy and information display.

The meeting significantly clarified the general vision and expectations of the stakeholders. A key outcome of the discussion was learning that the project should not be constrained by current technological limitations. Instead, the focus should remain on user's benefits, practical use cases and conceptual solutions. The stakeholders highlighted the importance of exploring multiple scenarios, for instance, when connectivity is available versus when it is not, and designing solutions that remain usable in both contexts.

From the meeting, a number of important themes emerged:

- **Translation:** Real time translation was considered a valuable feature, but stakeholders acknowledged the associated challenges of accuracy and trustworthiness. Maintaining an original language copy alongside translated text was suggested as a potential safeguard.
- **User groups:** Stakeholders confirmed that different user groups will have varying needs, with a particular focus on travelers, individuals with chronic conditions and healthcare providers.
- **Privacy and security:** While privacy remains a critical concern, stakeholders stressed that it should be treated within the same general framework as other secure services (e.g., banking). Data would primarily remain with government institutions and only be accessed through secure links and electronic identification, with users retaining control over which information to share.
- **Information selection:** A recurring theme was the importance of allowing users to choose what information to include in their medical summaries. This ensures relevance while addressing sensitive issues such as mental health history or misdiagnoses.
- **Trustworthiness:** The issue of certifying and validating information was raised as an important challenge. Although no standard solution currently exists, possibilities such as electronic signatures or doctor-certified notes were mentioned as areas worth exploring.

Overall, the meeting highlighted the project's focus on conceptual exploration and user centered design. The stakeholders encouraged the team to think creatively, keep solutions simple, and ground proposals in real world use cases. This meeting provided the team with a clearer direction and reduced earlier uncertainties.

2.5 Interviews with users

Each team member interviewed one individual that belongs to our established user groups, with a total of five interviews being conducted for Project B. A total of three men and two women with an age between 21 to 46 were interviewed. Also, these individuals have different digital backgrounds and education levels. Each interviewee provided perspective on the value and challenges of carrying a medical summary when traveling abroad.

2.5.1 Summary

Interview 1 A student with no prior experience in using electronic medical records but is supportive of the idea. Believes that the feature could be *"very helpful for people that want to carry it in an emergency situation"*, and would personally use it if given the opportunity.

Interview 2 An HR professional who is familiar with *Heilsuvera*. She perceived the idea of a medical summary as an *"incredibly helpful"* idea.

Interview 3 A master's student who has some experience with electronic records and supports the application if it was free and certified. Values intuitive, jargon-free design, instant emergency access and suggested security features such as anonymization features and automatic protection of sensitive personal data.

Interview 4 A general surgeon who has treated international patients and strongly supports the idea, stressing that quick access to reliable medical summaries would *"decrease any risks and complications in the long run"*

Interview 5 A user who is on medical leave who has experience carrying medical notes while traveling. Interviewee valued both digital and printed access and expressed that *"having access to medical documents would be a sense of safety so my family would not have to explain it to the doctors"*

2.5.2 Analysis of Interview Findings

Overall, the interviews revealed broad support and positive attitudes toward the medical summary system. Interviewees widely agreed the feature would improve care in emergencies and reduce stress when traveling, with one of the interviewees noting that it would give her peace of mind.

Interviewees were also aligned on what essential information should be included in medical summaries: Allergies, medications, chronic conditions, past surgeries, blood type, and emergency contacts. For instance, interviewee 4 said: *"Name, date of birth, allergies, current medication/prescription, past surgeries and major condition..."*. Additionally, interviewee 3 added that organ donation status should be included. At the same time, there was a consensus between interviewees that irrelevant or sensitive information should be excluded such as religion, sexuality and minor illnesses.

Regarding formats and accessibility, the interviewees emphasized the importance of providing flexible options for accessing medical summaries in different formats. Interviewee 1 preferred a

downloaded file that was simple to open, while interviewee 2 suggested "*a QR code that links to a secure summary... [with] a short printed summary as backup*", which was agreed upon by interviewee 3. Meanwhile, interviewee 4 favored a concise summary page integrated with hospital systems, whereas interviewee 5 suggested an option for printed access, with "*a QR code [that] should definitely have an expiry date*". The key requirement that all interviewees expressed was speed and simplicity of access, particularly in emergencies. Lastly, one interview pointed out a disadvantage is if patient arrives unconscious and the clinicians are unaware about the medical summary, which is a factor to take into consideration.

Lastly, another critical factor that emerged was security and trust. Although most interviewees were personally comfortable with sharing their data, they acknowledged the sensitivity of the information and demanded safeguards. Interviewee 1 suggested that doctors who are receiving the medical information should get a personal code to open the app with the medical summary, while interviewee 2 emphasized security and reliability. Interviewee 3 emphasized the need for automatic data censorship, so that when sharing their medical summary only essential information is shown while sensitive personal details remain hidden, giving them full control over their privacy. Interviewee 4 emphasized that security must be strong yet simple, with encrypted, nationally verified summaries that can be quickly accessed without leaving data stored on the device. Lastly, interviewee 5 stressed that access should be strictly limited to medical professionals, with expiring links and authentication (e.g., *rafræn skilríki*) to prevent abuse or unauthorized access by Ísland.is staff.

In summary, the interviews demonstrate strong interest and support for a portable electronic medical summary integrated into *Ísland.is*. Participants recognized clear benefits in terms of safety, reliability, and efficiency during medical encounters abroad. These user insights provide valuable guidance for the design of our system, highlighting what information should be included and how it should be accessed and secured. The findings confirm the relevance of this project and inform important design decisions moving forward.

2.6 UX goals

When defining the UX goals for the project, each team member downloaded the UX goals template provided on *UCDSprint.com*. Then, each member took time to reflect on the project's needs, after which they individually selected three UX goals. These were put in a shared document accessible to the entire team.

Once the individual goals were collected, the team worked collaboratively to organize them. Goals that were identical or closely related were grouped together to avoid redundancy and highlight common priorities. For the remaining goals, the team discussed their relevance to the project. Through this process of open discussion, the group identified and agreed on three core UX goals that would serve as the primary focus moving forward. These UX goals were **security**; you feel a sense of security by having access to correct medical care abroad, **relief**; you feel free from tension or concerns related to medical concerns while traveling, **control**; you are in control of your health, have access to up to date medical records and can use them to get better care or diagnoses abroad (see Appendix E for the filled out template). The choice of UX goals also informed the final long-term goal of our medical summary system, as visible in our map (see AppendixC.2): "Better medical treatment outside Iceland so that travelers feel secure, relieved, and in control".

This allows the team connect the immediate UX priorities and the overall outcome of the project, creating a clear framework for system development.

3 Conclusion

This report shows the discovery phase of Project B, highlighting the potential of extending the Ísland.is application to provide travelers with secure and accessible medical summaries abroad. With as-is analysis, mapping, stakeholder meeting, interviews and UX goals, the team identified the needs and challenges of implementing such a system.

The interviews revealed consistent themes across the two main user groups: travelers and healthcare professionals, each represented by participants from diverse backgrounds. Travelers emphasized the importance of safety, peace of mind, and convenience when accessing their medical records abroad while healthcare professionals stressed the need for clear, concise, and trustworthy summaries that could be quickly interpreted in emergencies. Both groups highlighted the balance between accessibility and security as crucial for the application.

Stakeholders pointed out that the project should focus on user benefits rather than technical limits, highlighting translation, privacy, and trust as key priorities for success. The refined mapping and UX goals; security, relief and control, established a clear framework for future development.

This discovery process has laid a solid foundation for moving forward into the design and prototyping phases. The results strongly suggest that a well designed medical summary feature could significantly enhance patient safety, improve healthcare delivery abroad, and provide Icelandic travelers with peace of mind when facing medical needs outside their home country.

A Interview Questions

A.1 Interview Guide for Traveler

We are conducting interviews as part of a project in the course Software Project Management – 2025.

Thank you for giving me the opportunity of interviewing you. I will write down what you say and I will use the results of the interview in the course. I will not write your name on the results, so it will not be possible to see who it is that was interviewed. I will keep anonymity. I presume the interview will not take longer than 30 minutes.

Is there something you would like to ask about before we begin?

The project is about extending the **Ísland.is application** with a new feature. The idea is that Icelandic citizens would be able to **carry a secure, offline copy of their medical summary** while traveling abroad.

This summary could be shown to doctors in case of an emergency and would ideally be **available in English** or translated “on the fly” into the local language of the country being visited.

The purpose of this interview is to understand how travelers feel about such a solution, what their needs would be, and what concerns they might have. Your answers will help us design a meaningful and user-friendly prototype.

1. To begin with, can you tell me about your background? (Education, nationality)
2. What is your current situation (job, studies, etc.)?
3. Do you have any experience with medical documents electronically?
4. Have you ever had a medical experience outside of Iceland?
 - (a) What was it like?
5. Have you ever had to carry medical documents or reports with you while traveling?
6. What are your opinions on having access to your electronic medical summary while traveling?
7. Would you personally use an app feature like this? Why or why not?
8. What functionalities would you like to see in such an app? What experience do you expect?
9. What would be important for you in an emergency?

10. What format would be most useful to present the information (e.g., printed, downloaded file, short summary page, QR code linking to a secure document, on-screen)? Why?
11. What information do you think must be included in the medical summary, and what information should not be included?
12. What are your thoughts on the security aspect since this is highly confidential information?
13. What would make you feel more secure when using such an app?
14. Before we conclude, is there anything else you would like to suggest or mention that we haven't asked about?

Thank you for your time and cooperation.

A.2 Interview Guide for Healthcare Professionals

We are conducting interviews as part of a project in the course Software Project Management – 2025.

Thank you for giving me the opportunity to interview you. I will write down what you say and I will use the results of the interview in the course. I will not write your name on the results, so it will not be possible to see who it is that was interviewed. I will keep anonymity. I presume the interview will not take longer than 30 minutes.

Is there something you would like to ask about before we begin?

The project is about extending the **Ísland.is application** with a new feature. The idea is that Icelandic citizens would be able to **carry a secure, offline copy of their medical summary** while traveling abroad.

This summary could be shown to doctors in case of an emergency and would ideally be **available in English** or translated “on the fly” into the local language. ***Purpose:*** The purpose of this interview is to understand the needs of healthcare professionals who might receive patients from abroad, to gain perspective on what healthcare professionals need if Icelanders were to travel abroad and need medical attention. so, we can design the solution in a way that is most useful for practice

1. To begin with, can you tell me about your background? (Education, nationality)
2. What is your current situation (job, studies, etc.)?
3. Have you had any experience with medial documents electronically?
 - (a) If so what are they?
4. Have you treated international patients before? Can you share any experiences where a lack of medical information was a challenge?
5. Have you received medical documents from patients (e.g., printed reports, digital files)?
 - (a) What worked well, and what was difficult?
6. What is your opinion about patients carrying a digital medical summary that can be shown during emergencies?

7. Do you see this being useful in your day-to-day work? Why or why not?
8. Which information would be most important for you to see in a medical summary (e.g., allergies, medications, diagnoses, vaccinations, blood type)?
9. And, is there any information that would not be necessary or even complicate your work?
10. How would you like the information to be formatted (e.g., short summary page, structured categories, QR code linking to a secure document, printed version)?
 - How would you like the medical summary to be structured? (Short and concise or detailed?)
11. In emergency situations, how quickly would you expect to access and interpret such a summary?
12. What are your thoughts on the security of the data, are there any functions that must be done to gain the information?
13. Before we conclude, is there anything else you would like to suggest or mention that we haven't asked about?

Thank you for your time and cooperation.

B User Group Analysis Table

NAME of group	Icelandic travelers	Healthcare professionals/clinicians abroad
WHO background	Age: Any age (children with parents, adults, elderly)	Age: 22 - 70
	Gender: All genders	Gender: All genders
	Education: Varies; not medical	Education: Medical education (MD, nurse, specialist)
	Abilities/Disabilities: All scope; may include chronic conditions, congenital conditions, disabilities, limited digital literacy	Abilities/Disabilities: Preferably no disabilities that limits professional role
	Computer skills: from none to advanced smartphone use	Computer skills: Familiar with Electronic health records
	Number: Potentially thousands of Icelandic citizens and residents travelling yearly	Number: Countless clinicians and healthcare professionals worldwide who might treat Icelandic citizens or residents
WHY main goals	To carry a reliable summary of health records for emergencies abroad	To have a reliable summary of health records for emergencies for Icelandic travelers
WHAT equipment	Smartphone with Ísland.is app or printed version	Clinics, hospitals and emergency room abroad
WHERE environment	When travelling abroad	Doctors office not in Iceland, hospital not in Iceland
WHEN usage of system	How often: When in need of emergency while travelling	How often: When an Icelandic patient is in need of medical care abroad
	For how long each time: Duration of medical encounters	For how long each time: Duration of medical encounters
	Skills: Ability to open app and present the qr code/ pdf, understand translation	Skills: Ability to read and understand medical records
HOW Important	To gain a better care management from the healthcare and their understanding on the personal condition (if present)	To gain a better assessment on patient and factor in medical history of the patient in its treatment

Figure B.1: A completed User Group Analysis Table from UCDSprint, including the user groups of Icelandic travelers and Healthcare Professionals/Clinicians abroad.

C Mappings

C.1 Original Mapping

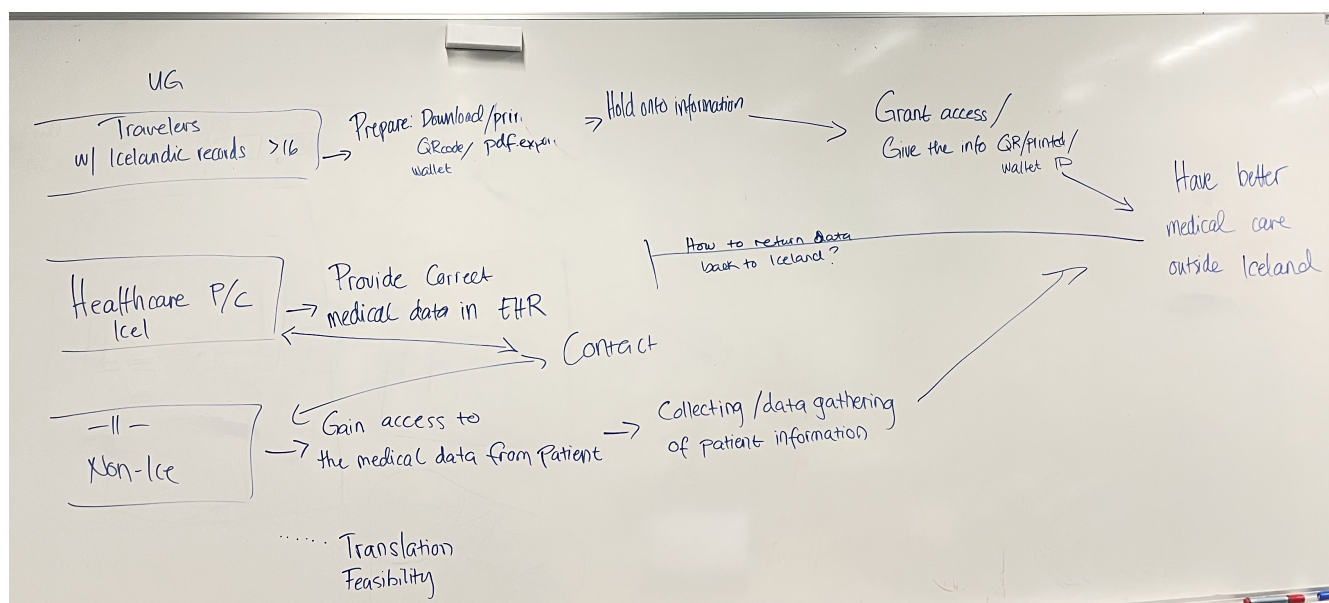


Figure C.1: Original Mapping on a whiteboard mapping three users groups (Travelers, Healthcare Professional/Clinicians in Iceland and Healthcare Professional/Clinicians abroad) to the final goal (Have better medical care outside Iceland)

C.2 Current Mapping

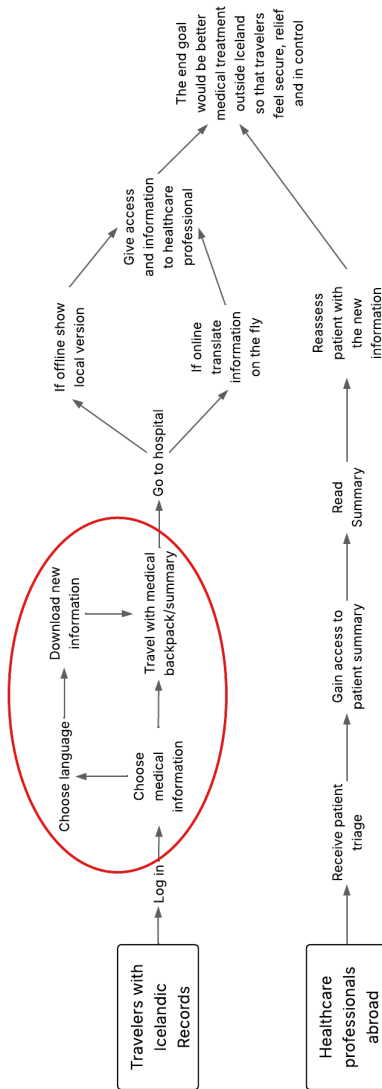


Figure C.2: Current Mapping from Two user groups (Travelers and Healthcare Professional/Clinicians abroad) to the end goal (Better medical treatment outside Iceland so that travelers feel secure, relief and in control). The red circle shows the target we chose for the rest of the sprint. It highlights an option for the traveler's preparation loop. The traveler could prepare beforehand and choose the language (of the country traveling to) and download new information. The traveler could also skip those two steps and travel with the current medical summary.

D Interviews

The answers may be rearranged to fit the questions in the interview guide, as some questions were answered during discussion of other questions.

D.1 Interview Transcript 1 (23, female)

Q: To begin with, can you tell me about your background? (Education, nationality)

A: I am 23 years old and live in the Netherlands.

Q: What is your current situation (job, studies, etc.)?

A: Currently, I am in the third year of my bachelor Nutrition & Health. Next to that I also have a part-time job as a waitress.

Q: Do you have any experience with medical documents electronically?

A: I don't.

Q: Have you ever had a medical experience outside of Iceland?

A: I had surgery twice in the Netherlands.

Q: What was it like?

A: From the first time, I don't remember that much because I was very young. From the second time I remember that the hospital staff took good care of me, which made it a good experience.

Q: Have you ever had to carry medical documents or reports with you while traveling?

A: No.

Q: What are your opinions on having access to your electronic medical summary while traveling?

A: I don't see a problem with it. I think it can be very helpful for people that want to carry it in an emergency situation.

Q: Would you personally use an app feature like this? Why or why not?

A: I would for sure. Like I said, it can be helpful in emergency situations, so why not? I can imagine that some people wouldn't want to because of privacy reasons, but that doesn't concern me at all.

Q: What functionalities would you like to see in such an app? What experience do you expect?

A: I think I would want a list of medications that you take or took, a record of your medical history, and a list with wishes about, for example, receiving CPR.

Q: What would be important for you in an emergency? What experience do you expect?

A: It would be important for me that I can be helped in the best possible way and that the medical

staff knows my medical history. I expect a good experience with the app. I can't really think of any negative sides to it for me. The only problem could be that medical staff doesn't know you have the app when you're unconscious.

Q: What format would be most useful to present the information (e.g., printed, downloaded file, short summary page, QR code linking to a secure document, on-screen)? Why?

A: For me a downloaded file would work best so that it is not complicated to open when it's needed.

Q: What information do you think must be included in the medical summary, and what information should not be included?

A: Medical history, medicines taken, preferences regarding e.g. receiving CPR. What should not be included: maybe things like religion or sexuality.

Q: What are your thoughts on the security aspect since this is highly confidential information?

A: I think getting such a file should be at own risk, but maybe that's easy for me to say, since my file isn't that interesting. So maybe after all, the file should be secured so that not everyone can open it. Maybe doctors should get a personal code they can enter in order to open the app, or something like that?

Q: What would make you feel more secure when using such an app?

A: I already feel quite secure with it, but maybe a login with social security number would make it feel more safe for me.

Q: Before we conclude, is there anything else you would like to suggest or mention that we haven't asked about?

A: No, not really.

D.2 Interview Transcript 2 (46, female)

Q: First, can you tell me about your background, education and current situation (job, studies, etc.)?

A: I was born and raised in Reykjavík, Iceland. I studied business at the University of Iceland and later specialized in human resources.

Q: What is your current situation (job, studies, etc.)?

A: I currently work in HR at Play Air.

Q: Do you have any experience with medical documents electronically?

A: Yes, in Iceland we have online access to our medical records through Heilsuvera. I've used it to view lab results, prescriptions, and doctor notes.

Q: Have you ever had a medical experience outside of Iceland?

A: No.

Q: Have you ever had to carry medical documents or reports with you while traveling?

A: No.

Q: What are your opinions on having access to your electronic medical summary while traveling?

A: I think it would be incredibly helpful. Doctors abroad could get a clearer picture quickly, and it would reduce the stress of remembering every detail.

Q: Would you personally use an app feature like this? Why or why not?

A: Yes, definitely. It would be convenient and give me peace of mind. If it's secure and reliable, it would be a great tool.

Q: What functionalities would you like to see in such an app? What experience do you expect?

A: A simple summary of key medical history (allergies, conditions, medications). Emergency contacts. Translations into the local language if possible. A way for doctors abroad to securely upload notes back to my Icelandic record.

Q: What would be important for you in an emergency?

A: Quick access without needing passwords or internet. Also, ensuring paramedics can easily see allergies or critical conditions.

Q: What format would be most useful to present the information (e.g., printed, downloaded file, short summary page, QR code linking to a secure document, on-screen)? Why?

A: An app would probably be best but with no internet it could fail. So maybe a QR code that links to a secure summary would be best. A short printed summary as backup would also be useful in case of no internet.

Q: What information do you think must be included in the medical summary, and what information should not be included?

A: Must include: allergies, medications, chronic conditions, blood type, emergency contacts, and vaccination status.

Q: What are your thoughts on the security aspect since this is highly confidential information?

A: I wouldn't want my medical history exposed. I'm not sure how, but I would want it to be very secure. Obviously if I print it out it is very easy to be exposed, but then maybe I could choose what is on the printed summary.

Q: What would make you feel more secure when using such an app?

A: Authentication.

Q: Is there something you would like to add?

A: No.

D.3 Interview Transcript 3 (24, male)

Q: First, can you tell me about your background, education and current situation (job, studies, etc.)?

A: I'm 24 years old and I'm from Italy. I studied computer science as an undergraduate.

Q: What is your current situation (job, studies, etc.)?

A: I am a Computer Science Master's student.

Q: Do you have any experience with medical documents electronically?

A: I've had some experience with electronic medical documents, mainly with things like blood test results and medical prescriptions.

Q: Have you ever had a medical experience outside of Iceland?

A: I haven't had any personal medical experiences abroad.

Q: Have you ever had to carry medical documents or reports with you while traveling?

A: Yes, I've had to carry physical medical documents with me while traveling. I did this primarily to avoid any issues at customs when carrying certain medications, like cardioaspirin. It can also be useful to have documents on hand, as some pharmacies might be a bit difficult about a young person buying specific medications without a clear reason.

Q: What are your opinions on having access to your electronic medical summary while traveling?

A: I see the importance and potential of having access to an electronic medical summary while traveling. It's a great idea, but only if it's a free service. Travel costs and health insurance can already be expensive, and adding another fee for this service wouldn't be appealing. If it was a free service offered by my home country, then I'd definitely use it. The most crucial part is that the summary needs to be officially recognized and trusted by doctors in other countries; otherwise, it would be useless.

Q: Would you personally use an app feature like this? Why or why not?

A: Yes, if the app is free and the summary has some sort of certification. If it meets these two conditions, its usefulness is unquestionable.

Q: What functionalities would you like to see in such an app? What experience do you expect?

A: For me, the app would need to be very intuitive and easy to use. I'd expect to find the information I need with just a few clicks. The app should use simple, clear language and avoid overly specific medical jargon that's hard for a non-expert to understand. If there's a difficult word, a tooltip could provide more detail. Basically, the app should be "idiot-proof" and straightforward.

Q: What would be important for you in an emergency?

A: In an emergency, speed of access is everything. I'd need to view critical information instantly, without dealing with complex menus or passwords. The data must also be certified and reliable. This allows a doctor to trust the information immediately and proceed straight to diagnosis and treatment, saving precious time by skipping redundant initial tests.

Q: What format would be most useful to present the information (e.g., printed, downloaded file, short summary page, QR code linking to a secure document, on-screen)? Why?

A: I think a PDF is the most useful format because it's universal and easy to share via email or messaging. A QR code is also very helpful, as it provides a quick, scannable link to the secure document. The ideal solution would be a combination of a QR code that links to a PDF.

Q: What information do you think must be included in the medical summary, and what information should not be included?

A: For me Allergies, Intolerances, Blood type, Organ donation status , Other detailed medical history could potentially be excluded to keep the summary concise and focused on what's most critical in an emergency.

Q: What are your thoughts on the security aspect since this is highly confidential information?

A: Security is a major concern since this is highly confidential information. If a doctor needs to download the file, I would prefer that the app automatically anonymizes some of my personal data. This would protect my privacy, especially since I might not know the local privacy laws wherever I am.

Q: What would make you feel more secure when using such an app?

A: To feel more secure, I would want an app with an automatic data censorship feature. When I choose to share my summary with a medical professional, the app should automatically hide

certain personal details before the document is downloaded or viewed. This would give me full control over my most private information.

Q: Is there something you would like to add?

A: No, there is nothing else I would like to suggest.

D.4 Interview Transcript 4 (33, male)

Q: First, can you tell me about your background, education and current situation (job, studies, etc.)?

A: I am a 33 year old male, I am from Iceland, I graduated in Medicine from University in Iceland. I recently came back from Sweden from my short residency program.

Q: What is your current situation (job, studies, etc.)?

A: I currently work as a general surgeon and am currently working to specialize in urology.

Q: Do you have any experience with medical documents electronically?

A: Yes, in Sweden and Iceland, both country uses electronic health records and I have used them daily while working.

Q: Have you treated international patients before? Can you share any experiences where a lack of medical information was a challenge?

A: Yes. One tourist with abdominal pain couldn't explain his surgical history, which delayed treatment until we confirmed it later. Another had unmarked medication with no records — we had to verify it manually, which took time.

Q: Have you received medical documents from patients (e.g., printed reports, digital files)? What worked well, and what was difficult?

A: Printed documents in English worked best when patient had it. What was difficult were long, handwritten notes or poorly translated documents. Burned CD with imagery testing results is also sometimes with patients especially when being transferred after being in a hospital abroad. That is an extra work for example.

Q: What is your opinion about patients carrying a digital medical summary that can be shown during emergencies?

A: I am in favor of the idea, I think it would reduce time at the emergency room for example. Emergencies don't wait for paperwork. Having a summary that provides important information about a patient's medical history that can affect the patient's condition will decrease any risks and complications in the long run. It can save time and reduce errors, especially with complex medical histories.

Q: Do you see this being useful in your day-to-day work? Why or why not?

A: Yes, it is valuable for patient safety. If a patient could quickly provide their medical history in a format that can be easily understood, then we can easily factor it in the equation and whether their emergency or medical attention is related to their medical condition.

Q: Which information would be most important for you to see in a medical summary (e.g., allergies, medications, diagnoses, vaccinations, blood type)?

A: Name, date of birth, allergies, current medication/prescription, past surgeries and major

conditions, blood type (if available), emergency contact, immunizations. Irrelevant information would be sensitive psychiatric notes and minor illnesses.

Q: And, is there any information that would not be necessary or even complicate your work?

A: Minor illnesses like I mentioned before. I also think long detailed history in the summary.

Q: How would you like the information to be formatted (e.g., short summary page, structured categories, QR code linking to a secure document, printed version)?

A: I would like it to be short, in like a printable summary page or something that can be worked with in our own health record data. I think it would be convenient if the information can be popped up in our computer environment. A short version and, if required, access to detailed information — but that should not be the priority.

Q: In emergency situations, how quickly would you expect to access and interpret such a summary?

A: I would like to have access to it within seconds — ideally no more than a minute to understand the essentials.

Q: What are your thoughts on the security of the data, are there any functions that must be done to gain the information?

A: Security is crucial — patients won't trust it if the app is not secure. But the app should be simple to use in a critical or stressful situation. Summary should be encrypted and verified by the national healthcare. How? Not sure. Locked and accessed with authentication or pre-downloaded with date of when it was downloaded. No data memory.

Q: Before we conclude, is there anything else you would like to suggest or mention that we haven't asked about?

A: The app should be intuitive and maybe allow patients to delegate access to a family member. Also, I would like as a medical staff we should not need training to read or learn how to get the information — it should be very simple, clear, and universal.

D.5 Interview Transcript 5 (21, Male)

Q: First, can you tell me about your background, education and current situation (job, studies, etc.)?

A: I'm a 21 year old person that has a student'spróf.

Q: What is your current situation (job, studies, etc.)?

A: I am on medical leave.

Q: Do you have any experience with medical documents electronically?

A: No.

Q: Have you ever had a medical experience outside of Iceland?

A: No, only in Iceland.

Q: Have you ever had to carry medical documents or reports with you while traveling?

A: Yes, medical exemptions for traveling with drugs, [and] Doctors Note.

Q: What are your opinions on having access to your electronic medical summary while traveling?

A: I would like it. I am at risk of having a medical emergency while traveling so having access to them [medical documents] would be a sense of safety so my family wouldn't have to explain it

to the doctors, so I could just have access to it or even use my access to it to print it and travel with it, so I don't have to go to the doctor every time, not just to have prescriptions but in case something happens and I have to go to the hospital. Yeah, it would be very nice to have a summary to travel with, even just to have access to it when I'm conscious, or also be able to travel with it in my pocket and printed out or something similar.

Q: Would you personally use an app feature like this? Why or why not?

A: Yes I would definitely use it.

Q: What functionalities would you like to see in such an app? What experience do you expect?

A: A print-out button or to be able to save it as pdf, but it should have a date when you print it out, when it was printed out or when the summary was last updated, and when it was printed out so the doctor can see [that] nothing has happened in the last year, but he printed this out just before he bought tickets—; that would be a big one. Also so [that] you could send it to a doctor if you are in a hospital and they could send it to the next doctor that needs it in their hospital system or something similar, maybe even so [that] you can designate **umboðsaðili**[agent], so a family member can access your medical history in case something happens to you while you are outside of the country; these are the two important ones.

Q: What would be important for you in an emergency?

A: Having a section of immediate concerns such as if someone has a condition or a heart condition, if they lose muscle function or pass out they are at risk of dying. Something very serious that should be known instantly to doctors, so they don't have to look through the entire history, just a summary that says this problem, this problem, this problem, and then they can access this is not related, we don't have to read through the whole document. Maybe a section with threats and major conditions.

Q: What format would be most useful to present the information (e.g., printed, downloaded file, short summary page, QR code linking to a secure document, on-screen)? Why?

A: Definitely printed and QR code, but the QR code should definitely have an expiry date so [that] someone cannot take a picture of it and access your health information forever. [Access controls] would be helpful if you are conscious, but if something happens and you are unconscious it's a risk. So maybe you could allow a QR code to work 30 days or some set time where that QR code would be generated, and then a doctor could or you could have it printed out and have it with you or on your phone as the emergency contact thingy as like a link. For emergencies, you definitely need someone to be able to access it without your consent, so printed out is one of the bigger factors, but also so you could send it or share it without having to print it out, perhaps to the email of the doctor you are working with or the medical staff at the hospital in a foreign country.

Q: What information do you think must be included in the medical summary, and what information should not be included?

A: Information that doesn't have a risk factor for other conditions, so if you had say a foot fungus infection or athlete's foot maybe it should not be in the summary or something similar unless it's repeating—if you went to **heilsugæsla** the clinic once, or say you had something that you cannot get again [hlaupabóla], it should not be in your summary that you had **hlaupabóla**[varicella] when you were four. But maybe include if you have not had it or a basic checklist for has had it or not, and more detailed descriptions could be later. [User checklist question] Yeah, I definitely would like something like that but not sure how it would be best to implement it. Being able to select what you want to show especially if you are conscious and see they only need information about this one

thing, select that and send it to them, it would be nice to have.

Q: What are your thoughts on the security aspect since this is highly confidential information?

A: Yeah, that's a tough one to balance since security is... yeah, you have to think about it. Say if someone steals your paper or something it's the same as if you had manually typed it out. But if you have a way to share it with a link or like that, it should definitely expire so someone can't repeatedly access it. Or say if it's stored in someone else's system with a link and other persons can't access it [the link] it should expire after a certain amount of days. But it's a balancing act you have to think about.

Q: What would make you feel more secure when using such an app?

A: That it couldn't be abused like the doctors lookup thing where random doctors have been looking up your name and seeing your health conditions. Staff working with island.is should not be able to access your summary—if you are not a medical professional you should not, if you just work with the database you should not be able to access. Having a way to lock it out or you have to have **rafræn skilríki**[Digital Identity] or island.is password to access it.

E User Experience Goals

	Experience	Description
Team member 1	Confidence	To experience mental or moral strength to withstand or cope with the situation
	Relaxation	To enjoy the recent removal of stress or discomfort
	Security	To experience safe and in control of your life rather than feeling uncertain and threatened by your circumstances
Team member 2	Security	You have security in that you can get access to correct medical care abroad
	Relief	You fell free from tension or concerns related to medical concerns while traveling
	Control	You are in control of you health/you have access to up to date medical records and can give them to get better care/diagnosis abroad
Team member 3	Competence	To experience that you are very capable and effective in your actions rather than feeling incompetent or ineffective
	Autonomy	To experience you are the cause of your own actions rather than feeling that external forces or pressure are the cause of your action
	Relief	You fell free from tension or concerns related to medical concerns while traveling
Team member 4	Relief	To enjoy a calm state of being free from mental or physical tension or concern
	Autonomy	To experience you are the cause of your own actions rather than feeling that external forces or pressure are the cause of your action
	Security	To experience safe and in control of your life rather than feeling uncertain and threatened by your circumstances.
Team member 5	Security	You have security in that you can get access to correct medical care abroad
	Control	You are in control of you health/you have access to up to date medical records and can give them to get better care/diagnosis abroad
	Relief	To enjoy a calm state of being free from mental or physical tension or concern
Final UX goals	Security	You have security in that you can get access to correct medical care abroad
	Relief	You fell free from tension or concerns related to medical concerns while traveling
	Control	You are in control of you health/you have access to up to date medical records and can give them to get better care/diagnosis abroad

Figure E.1: Table showing what three UX goals each team member selected with associated description fitting to the scenario, also including Final three UX goals selected by the team members (Security, Relief and Control).